MATH1083: Calculus II Midterm Solution

1. Determine whether the series is divergent, absolutely convergent or conditionally convergent.

$$(a)\sum_{n=1}^{\infty}\frac{3n+1}{2n-5}$$
$$(b)\sum_{n=2}^{\infty}\frac{(-1)^n}{\ln n}$$
$$(c)\sum_{n=1}^{\infty}\frac{10^n}{n!}$$
$$(d)\sum_{n=1}^{\infty}\left(\frac{3n+2}{7n-1}\right)^n$$

Solution:

(a) Compute the limit of the general term:

$$\lim_{n\to\infty}\frac{3n+1}{2n-5}=\lim_{n\to\infty}\frac{3+\frac{1}{n}}{2-\frac{5}{n}}=\frac{3}{2}\neq 0.$$

By the Test for Divergence, since the limit is not zero, the series diverges.

(b) Absolute Convergence: Check $\sum_{n=2}^{\infty}\frac{1}{\ln n}$. Compare with the divergent series $\sum_{n=2}^{\infty}\frac{1}{n}$. For $n\geq 3$, $\frac{1}{\ln n}>\frac{1}{n}$. By the Comparison Test, $\sum_{n=2}^{\infty}\frac{1}{\ln n}$ diverges. Thus, the original series does not converge absolutely.

Conditional Convergence: Check the alternating series $\sum_{n=2}^{\infty}\frac{(-1)^n}{\ln n}$. The sequence $\frac{1}{\ln n}$ is positive, decreasing for $n\geq 2$, and $\lim_{n\to\infty}\frac{1}{\ln n}=0$. By the Alternating Series Test, the series converges.

Conclusion: The series converges conditionally.

(c) Compute the limit:

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\frac{10^{n+1}}{(n+1)!}\cdot\frac{n!}{10^n}=\lim_{n\to\infty}\frac{10}{n+1}=0.$$

Since $0<1$, by the Ratio Test, the series converges absolutely.

(d) Compute the limit:

$$\lim_{n\to\infty}\sqrt[n]{|a_n|}=\lim_{n\to\infty}\frac{3n+2}{7n-1}=\frac{3}{7}$$

Since $\frac{3}{7}<1$, by the Root Test, the series converges absolutely.

2. Find the radius of convergence and interval of convergence of the series

$$\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{\sqrt{n+1}}$$

Solution: Let $a_n = (-3)^n x^n / \sqrt{n+1}$. Then

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-3)^{n+1}x^{n+1}}{\sqrt{n+2}} \cdot \frac{\sqrt{n+1}}{(-3)^n x^n}\right| = \left|-3x\sqrt{\frac{n+1}{n+2}}\right|$$
$$= 3\sqrt{\frac{1+(1/n)}{1+(2/n)}}|x| \to 3|x| \quad \text{as } n \to \infty$$

By the Ratio Test, the given series converges if $3|x| < 1$ and diverges if $3|x| > 1$. Thus it converges if $|x| < \frac{1}{3}$ and diverges if $|x| > \frac{1}{3}$. This means that the radius of convergence is $R = \frac{1}{3}$.

We know the series converges in the interval $\left(-\frac{1}{3}, \frac{1}{3}\right)$, but we must now test for convergence at the endpoints of this interval. If $x = -\frac{1}{3}$, the series becomes

$$\sum_{n=0}^{\infty} \frac{(-3)^n \left(-\frac{1}{3}\right)^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{1}{\sqrt{n+1}} = \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \cdots$$

which diverges. (Use the Integral Test or simply observe that it is a p-series with $p = \frac{1}{2} < 1$.) If $x = \frac{1}{3}$, the series is

$$\sum_{n=0}^{\infty} \frac{(-3)^n \left(\frac{1}{3}\right)^n}{\sqrt{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$$

which converges by the Alternating Series Test. Therefore the given power series converges when $-\frac{1}{3} < x < \frac{1}{3}$, so the interval of convergence is $\left(-\frac{1}{3}, \frac{1}{3}\right]$.

3. (a) Use series to evaluate the limit

$$\lim_{x \to 0} \frac{\sqrt{1+x} - 1 - \frac{1}{2}x}{x^2}$$

(b) Use series to approximate the definite integral to within the indicated accuracy.

$$\int_0^1 \sin\left(x^4\right) dx \quad \text{(four decimal places)}$$

Solution:

(a)

$$\lim_{x \to 0} \frac{\sqrt{1+x} - 1 - \frac{1}{2}x}{x^2} = \lim_{x \to 0} \frac{\left(1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots\right) - 1 - \frac{1}{2}x}{x^2} = \lim_{x \to 0} \frac{-\frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots}{x^2}$$
$$= \lim_{x \to 0} \left(-\frac{1}{8} + \frac{1}{16}x - \cdots\right) = -\frac{1}{8}$$

since power series are continuous functions.

(b) $\sin x = \sum_{n=0}^{\infty}(-1)^n \frac{x^{2n+1}}{(2n+1)!}$ for all x, so $\sin(x^4) = \sum_{n=0}^{\infty}(-1)^n \frac{x^{8n+4}}{(2n+1)!}$ for all x and $\int \sin(x^4)\,dx = C + \sum_{n=0}^{\infty}(-1)^n \frac{x^{8n+5}}{(2n+1)!(8n+5)}$. Thus, $\int_0^1 \sin(x^4)\,dx = \sum_{n=0}^{\infty}(-1)^n \frac{1}{(2n+1)!(8n+5)} = \frac{1}{1!\cdot 5} - \frac{1}{3!\cdot 13} + \frac{1}{5!\cdot 21} - \frac{1}{7!\cdot 29} + \cdots$. Now $\frac{1}{1!\cdot 5} - \frac{1}{3!\cdot 13} + \frac{1}{5!\cdot 21} \approx 0.1876$ and subtracting $\frac{1}{7!\cdot 29} \approx 6.84 \times 10^{-6}$ does not affect the fourth decimal place, so $\int_0^1 \sin(x^4)\,dx \approx 0.1876$ by the Alternating Series Estimation Theorem.

4. Let $\mathbf{r}(t) = (\sin 2t, 3t, \cos 2t)$, $-\pi \le t \le \pi$.

   (a) Show that the velocity and acceleration vectors are always perpendicular.

   (b) Is there a time t for which $\mathbf{r}(t)$ and the velocity vector are perpendicular? If so, find all such values of t.

   Solution:

   (a)

$$\begin{aligned} \mathbf{v}(t) &= (2\cos 2t, 3, -2\sin 2t) \\ \mathbf{a}(t) &= (-4\sin 2t, 0, -4\cos 2t) \\ \mathbf{v}(t)\cdot\mathbf{a}(t) &= 0 \end{aligned}$$

   (b) $\mathbf{r}(t)\cdot\mathbf{v}(t) = 2\sin 2t\cos 2t + 9t - 2\sin 2t\cos 2t = 0$ implies that $t = 0$. So, the only time when $\mathbf{r}(t)$ and the velocity vector $\mathbf{v}(t)$ are perpendicular is $t = 0$.

5. Consider the sphere $x^2 + y^2 + z^2 = 9$. Find the equation of the plane tangent to this sphere at

   (a) (3,0,0).

   (b) (2,2,1).

   Solution:

   (a) $x = 3$

   (b) Near $(2, 2, 1)$ the value of z is positive, so we can rewrite the equation $x^2+y^2+z^2 = 9$ as

$$z = \sqrt{9 - x^2 - y^2}.$$

   Therefore,

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{1}{2}\frac{(-2x)}{\sqrt{9-x^2-y^2}} = \frac{-x}{\sqrt{9-x^2-y^2}} \\ \frac{\partial z}{\partial y} &= \frac{-y}{\sqrt{9-x^2-y^2}} \end{aligned}$$

   At $(x, y) = (2, 2)$, we have

$$\frac{\partial z}{\partial x}(2,2) = -2, \quad \frac{\partial z}{\partial y}(2,2) = -2.$$

So a tangent vector at $(2, 2, 1)$ is

$$\mathbf{n} = (-2, -2, -1).$$

So the tangent plane is

$$\mathbf{n} \cdot (\mathbf{x} - (2, 2, 1)) = 0.$$

That is $2x + 2y + z = 9$.

6. Prove that for positive real numbers a, b, c,

$$(a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9.$$

When does equality hold?

Hint: Use the Cauchy-Schwarz Inequality.

Proof:

Let's choose:

$$\mathbf{x} = (\sqrt{a}, \sqrt{b}, \sqrt{c}), \quad \mathbf{y} = \left(\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}}\right).$$

Then, the dot product is:

$$\mathbf{x} \cdot \mathbf{y} = \sqrt{a} \cdot \frac{1}{\sqrt{a}} + \sqrt{b} \cdot \frac{1}{\sqrt{b}} + \sqrt{c} \cdot \frac{1}{\sqrt{c}} = 1 + 1 + 1 = 3.$$

The norms are:

$$\|\mathbf{x}\| = \sqrt{a + b + c}, \quad \|\mathbf{y}\| = \sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}.$$

By Cauchy-Schwarz:

$$(\mathbf{x} \cdot \mathbf{y})^2 \leq \|\mathbf{x}\|^2 \|\mathbf{y}\|^2 \implies 3^2 \leq (a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right).$$

Thus,

$$(a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9.$$

Equality holds in Cauchy-Schwarz when $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent, i.e.,

$$\sqrt{a} = \lambda \cdot \frac{1}{\sqrt{a}}, \quad \sqrt{b} = \lambda \cdot \frac{1}{\sqrt{b}}, \quad \sqrt{c} = \lambda \cdot \frac{1}{\sqrt{c}}.$$

This simplifies to $a = b = c$.

7. Let $f(x, y)$ be a function defined as:

$$f(x, y) = \begin{cases} \frac{x^3 y}{x^4 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Prove or disprove: The function $f(x, y)$ is continuous at the origin $(0, 0)$.

Hint: To prove continuity at a point for a function of several variables, you need to show that the limit of the function as it approaches that point is equal to the value of the function at that point. Polar coordinates can be used to help with the limit calculation.

Proof:

In the polar coordinate system, every point in $\mathbb{R}^2$ can be represented as $(r\cos\theta, r\sin\theta)$, where $r \geq 0$ is the distance from the origin and θ is the angle measured from the positive x-axis.

So, we substitute $x = r\cos\theta$ and $y = r\sin\theta$ into $f(x, y)$, which gives us:

$$f(r\cos\theta, r\sin\theta) = \frac{(r\cos\theta)^3(r\sin\theta)}{(r\cos\theta)^4 + (r\sin\theta)^2} = \frac{r^4\cos^3\theta\sin\theta}{r^4\cos^4\theta + r^2\sin^2\theta}$$

We can simplify this further by dividing the numerator and the denominator by r^4 (since $r \neq 0$), which gives:

$$f(r\cos\theta, r\sin\theta) = \frac{\cos^3\theta\sin\theta}{\cos^4\theta + \frac{\sin^2\theta}{r^2}}$$

As (x, y) approaches $(0, 0)$, r approaches 0. So, the limit of $f(x, y)$ as (x, y) approaches $(0, 0)$ is:

$$\lim_{r\to 0} \frac{\cos^3\theta\sin\theta}{\cos^4\theta + \frac{\sin^2\theta}{r^2}}$$

No matter what value θ takes, as r approaches 0, the term $\frac{\sin^2\theta}{r^2}$ goes to infinity, making the whole fraction go to 0. Thus, the limit of $f(x, y)$ as (x, y) approaches $(0, 0)$ is 0, which is equal to $f(0, 0)$.

Therefore, the function $f(x, y)$ is continuous at the origin $(0, 0)$.

8. (a) **Solution:** Observe that the partial derivatives

$$\frac{\partial f}{\partial x} = \frac{(x^2+y^2)\left(ye^{xy} - \sin x\right) - 2x\left(\cos x + e^{xy}\right)}{\left(x^2+y^2\right)^2}$$
$$\frac{\partial f}{\partial y} = \frac{(x^2+y^2)\,xe^{xy} - 2y\left(\cos x + e^{xy}\right)}{\left(x^2+y^2\right)^2}$$

are continuous except when $x = 0$ and $y = 0$, so this function is C^1 and hence differentiable at all points other than $(0, 0)$. (It is not defined at $(0, 0)$.)

(b) Let $f(x, y) = x^2 + 3y$. Find the linear approximation of $f(x, y)$ near $(2, 4)$, and use it to estimate $f(2.01, 4.03)$.

Solution:

The linear approximation (tangent plane) of f at $(a, b) = (2, 4)$ is:

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

Compute the partial derivatives:

$$f_x = 2x \Rightarrow f_x(2,4) = 4, \quad f_y = 3 \Rightarrow f_y(2,4) = 3.$$

Thus,

$$L(x,y) = (2^2 + 3 \cdot 4) + 4(x-2) + 3(y-4) = 16 + 4(x-2) + 3(y-4).$$

The estimate is:

$$f(2.01, 4.03) \approx L(2.01, 4.03) = 16 + 4(0.01) + 3(0.03) = 16 + 0.04 + 0.09 = 16.13.$$

(c) Let $f(x,y) = 4 + 3x + 2y + xy$ and $\mathbf{A} = (1,1)$. Find the directional derivatives $D_{\mathbf{V}} f(\mathbf{A})$ in the following directions:

$$\mathbf{V} = (1,0), \quad \mathbf{V} = \left(\frac{4}{5}, \frac{3}{5}\right).$$

Solution:
The directional derivative is given by:

$$D_{\mathbf{V}} f(\mathbf{A}) = \nabla f(\mathbf{A}) \cdot \mathbf{V}.$$

First, compute the gradient:

$$\nabla f = (f_x, f_y) = (3 + y, 2 + x) \Rightarrow \nabla f(1,1) = (4,3).$$

Now compute for each direction:

i. For $\mathbf{V} = (1,0)$:

$$D_{\mathbf{V}} f(\mathbf{A}) = (4,3) \cdot (1,0) = 4.$$

ii. For $\mathbf{V} = \left(\frac{4}{5}, \frac{3}{5}\right)$:

$$D_{\mathbf{V}} f(\mathbf{A}) = (4,3) \cdot \left(\frac{4}{5}, \frac{3}{5}\right) = \frac{16}{5} + \frac{9}{5} = 5.$$

9. Let $u(x,y) = x^2 - y^2$ and $v(x,y) = 2xy$.

(a) Show that $u_x = v_y$ and $u_y = -v_x$.
Solution:
Compute the partial derivatives:

$$u_x = 2x, \quad v_y = 2x \quad \Rightarrow u_x = v_y.$$

$$u_y = -2y, \quad v_x = 2y \quad \Rightarrow u_y = -v_x.$$

(b) Show that $u_{xx} + u_{yy} = 0$.
Solution:
Compute second derivatives:

$$u_{xx} = 2, \quad u_{yy} = -2 \quad \Rightarrow u_{xx} + u_{yy} = 0.$$

(c) Define $w(x,y) = u(u(x,y), v(x,y))$. Show that $w_{xx} + w_{yy} = 0$.
Solution: First, expand $w(x,y)$:

$$w(x,y) = u(u(x,y), v(x,y)) = u(x^2 - y^2, 2xy) = (x^2 - y^2)^2 - (2xy)^2.$$

Simplify:

$$w(x,y) = x^4 - 2x^2y^2 + y^4 - 4x^2y^2 = x^4 + y^4 - 6x^2y^2.$$

Now compute second derivatives:

$$w_x = 4x^3 - 12xy^2, \quad w_{xx} = 12x^2 - 12y^2,$$

$$w_y = 4y^3 - 12x^2y, \quad w_{yy} = 12y^2 - 12x^2.$$

Summing them:

$$w_{xx} + w_{yy} = (12x^2 - 12y^2) + (12y^2 - 12x^2) = 0.$$

(d) Suppose p, q, and r are C^2 functions such that

$$p_x = q_y, \quad p_y = -q_x, \quad r_{xx} + r_{yy} = 0.$$

Define $w(x,y) = r(p(x,y), q(x,y))$. Show that $w_{xx} + w_{yy} = 0$.
Solution: Using the chain rule, we compute:

$$w_x = r_p p_x + r_q q_x, \quad w_y = r_p p_y + r_q q_y.$$

Second derivatives:

$$\begin{aligned} w_{xx} &= (r_{pp}p_x + r_{pq}q_x) \cdot p_x + r_p p_{xx} + (r_{qp}p_x + r_{qq}q_x) \cdot q_x + r_q q_{xx} \\ &= r_{pp}p_x^2 + 2r_{pq}p_x q_x + r_{qq}q_x^2 + r_p p_{xx} + r_q q_{xx}, \end{aligned}$$

$$\begin{aligned} w_{yy} &= (r_{pp}p_y + r_{pq}q_y) \cdot p_y + r_p p_{yy} + (r_{qq}q_y + r_{qp}p_y) q_y + r_q q_{yy} \\ &= r_{pp}p_y^2 + 2r_{pq}p_y q_y + r_{qq}q_y^2 + r_p p_{yy} + r_q q_{yy}. \end{aligned}$$

Summing and using $p_x = q_y$, $p_y = -q_x$, and $r_{xx} + r_{yy} = 0$:

$$w_{xx}+w_{yy} = r_{pp}(p_x^2+p_y^2)+r_{qq}(q_x^2+q_y^2)+r_p(p_{xx}+p_{yy})+r_q(q_{xx}+q_{yy})+2r_{pq}(p_x q_x+p_y q_y).$$

Since p and q satisfy (from $p_x = q_y$, $p_y = -q_x$):

$$p_x q_x + p_y q_y = 0, \quad p_{xx} + p_{yy} = q_{yx} - q_{xy} = 0, \quad q_{xx} + q_{yy} = -p_{yx} + p_{xy} = 0.$$

Thus:

$$w_{xx} + w_{yy} = r_{pp}(p_x^2 + p_y^2) + r_{qq}(q_x^2 + q_y^2).$$

But $p_x^2 + p_y^2 = q_y^2 + q_x^2$, and since $r_{pp} + r_{qq} = 0$, we get:

$$w_{xx} + w_{yy} = (r_{pp} + r_{qq})(p_x^2 + p_y^2) = 0.$$